

Tokens, Attention, and Transformer Blocks

Large Language Models do not read text the same way we humans do; instead, text is first, broken down into **tokens**, which are the fundamental linguistic units (words or sub-word fragments) mapped to unique numerical identifiers. These identifiers are converted into continuous vectors called embeddings, which capture base semantic meanings. Next, these vectors enter the **Transformer blocks**, the modular structural units of the network stacked sequentially to process information. Within each block, the **Self-Attention** mechanism dynamically computes relationships between tokens by calculating how much "attention" each token should pay to every other token in the sequence. It achieves this by creating Query, Key, and Value vectors for each token, using dot-product mathematics to determine contextual weights, and blending the vectors accordingly. This allows the network to dynamically update a token's meaning based on its surrounding context (e.g., distinguishing "bank" in a financial context versus a geographical one).

As data flows deeper through the stacked Transformer blocks, the refined token vectors pass through feed-forward neural networks within each block to extract higher-level abstract features and logic. Each layer builds upon the context established by the previous one. At the end of the final block, the modified vectors reach the output layer, where a linear transformation and a softmax function project the mathematical values back into the model's vocabulary space. This generates a probability distribution across all possible tokens, allowing the model to accurately predict and output the most contextually appropriate next token in the sequence.

